Transmission Matrix Formulation for Anisotropic Slabs

1. Crystal Properties and Eigenvalue Problem

In an anisotropic electric and magnetic crystal, the modes satisfy the following eigenvalue problem

$$\left[\left(\vec{p}\times\overleftrightarrow{m}\right)\left(\vec{p}\times\overleftrightarrow{e}\right)+\overleftrightarrow{1}\right]\vec{D}_0=0 \tag{0.1}$$

Where $\vec{k}=k_0\vec{p}$, $\overleftrightarrow{m}=\overleftrightarrow{\mu}_r^{-1}$, $\overleftrightarrow{e}=\overleftrightarrow{\varepsilon}_r^{-1}$. For the scanner we enforce that $\vec{p}=p_x\hat{x}+p_z\hat{z}=\sin\theta\hat{x}+p_z\hat{z}$. The incident angle is set by the scanner while p_z is yet to be determined

In a source-free region, $\nabla\cdot D=0\rightarrow D_z=-\frac{p_x}{p_z}D_x$. Ultimately, this implies that only the x and y component of (1.1) are independent. Thus, our eigenvalue problem becomes

$$\left[\overleftrightarrow{G}+\overleftrightarrow{1}\right]\begin{bmatrix}D_x\\D_y\end{bmatrix}=0 \tag{0.2}$$

Let

$$\sigma_1=e_{xy}m_{xy}-e_xm_y$$
$$\sigma_2=2e_{\mathrm{zx}}m_y-e_{\mathrm{yz}}m_{\mathrm{xy}}-e_{\mathrm{xy}}m_{\mathrm{yz}}$$
$$\sigma_3=e_{\mathrm{yz}}m_{\mathrm{yz}}-e_zm_y$$
$$\sigma_4=e_ym_{\mathrm{xy}}-e_{\mathrm{xy}}m_y$$
$$\sigma_5=e_{\mathrm{yz}}m_y-e_ym_{\mathrm{yz}}$$
$$\sigma_6=e_xm_{\mathrm{xy}}-m_xe_{\mathrm{xy}}$$
$$\sigma_7=e_{\mathrm{yz}}m_x-e_xm_{\mathrm{yz}}+2e_{\mathrm{xy}}m_{\mathrm{zx}}-2e_{\mathrm{zx}}m_{\mathrm{xy}}$$
$$\sigma_8=e_zm_{\mathrm{xy}}-e_{\mathrm{xy}}m_z-2e_{\mathrm{yz}}m_{\mathrm{zx}}+2e_{\mathrm{zx}}m_{\mathrm{yz}}$$
$$\sigma_9=e_{\mathrm{yz}}m_z-e_zm_{\mathrm{yz}}$$
$$\sigma_{10}=e_{\mathrm{xy}}m_{\mathrm{xy}}-e_ym_x$$
$$\sigma_{11}=2e_ym_{\mathrm{zx}}-e_{\mathrm{yz}}m_{\mathrm{xy}}-e_{\mathrm{xy}}m_{\mathrm{yz}}$$
$$\sigma_{12}=e_{\mathrm{yz}}m_{\mathrm{yz}}-e_ym_z$$

Then

$$G_{xx}=\sigma_1p_z^{\,2}+\sigma_2\sin(\theta)p_z+\sigma_3\sin(\theta)^2 \tag{0.3}$$

$$G_{xy}=\sigma_4p_z^{\,2}+\sigma_5\sin(\theta)p_z \tag{0.4}$$

$$G_{yx}=\tfrac{1}{p_z}\left(\sigma_6p_z^{\,3}+\sigma_7\sin(\theta)p_z^{\,2}+\sigma_8\sin(\theta)^2p_z+\sigma_9\sin(\theta)^3\right) \tag{0.5}$$

$$G_{yy}=\sigma_{10}p_z^{\,2}+\sigma_{11}\sin(\theta)p_z+\sigma_{12}\sin(\theta)^2 \tag{0.6}$$

This allows our polynomial coefficients to be written

$$\begin{aligned}
A_0(\theta) &= (\sigma_3\sigma_{12} - \sigma_5\sigma_9)\sin(\theta)^4 + (\sigma_3 + \sigma_{12})\sin(\theta)^2 + 1 \\
A_1(\theta) &= (\sigma_2\sigma_{12} - \sigma_5\sigma_8 - \sigma_4\sigma_9 + \sigma_3\sigma_{11})\sin(\theta)^3 + (\sigma_2 + \sigma_{11})\sin(\theta) \\
A_2(\theta) &= (\sigma_1\sigma_{12} - \sigma_5\sigma_7 - \sigma_4\sigma_8 + \sigma_2\sigma_{11} + \sigma_3\sigma_{10})\sin(\theta)^2 + \sigma_1 + \sigma_{10} \\
A_3(\theta) &= (\sigma_1\sigma_{11} - \sigma_5\sigma_6 - \sigma_4\sigma_7 + \sigma_2\sigma_{10})\sin(\theta) \\
A_4(\theta) &= \sigma_1\sigma_{10} - \sigma_4\sigma_6
\end{aligned} \tag{0.7}$$

such that

$$0 = f(p_z) = A_0(\theta) + A_1(\theta)p_z + A_2(\theta)p_z^2 + A_3(\theta)p_z^3 + A_4(\theta)p_z^4 \tag{0.8}$$

Note, we have enforced reciprocity (**not** lossless) in that $\overleftrightarrow{m} = \overleftrightarrow{\mu}_r^{-1}$ and $\overleftrightarrow{e} = \overleftrightarrow{\varepsilon}_r^{-1}$ are assumed to be symmetric tensors. Therefore, we have 12 unknowns we need to solve for

$$\text{Unknowns: } m_x, m_y, m_z, m_{xy}, m_{yz}, m_{zx}, e_x, e_y, e_z, e_{xy}, e_{yz}, e_{zx}, \tag{0.9}$$

The relation between the magnetic and electric flux densities within the slab can be written in the spectral domain as

$$\vec{B} = \eta_0 (\vec{p} \times \vec{e}) \vec{D} \tag{0.10}$$

2. SPAM S-Parameter to Transmission Matrix

The scattering matrix measure by the SPAM system is in the h-v (horizontal/vertical) coordinate system (**make figure for this**) such that

$$\begin{bmatrix} E_{1h}^- \\ E_{1v}^- \\ E_{2h}^- \\ E_{2v}^- \end{bmatrix} = \begin{bmatrix} S_{hh}^{11} & S_{hv}^{11} & S_{hh}^{12} & S_{hv}^{12} \\ S_{vh}^{11} & S_{vv}^{11} & S_{vh}^{12} & S_{vv}^{12} \\ S_{hh}^{21} & S_{hv}^{21} & S_{hh}^{22} & S_{hv}^{22} \\ S_{vh}^{21} & S_{vv}^{21} & S_{vh}^{22} & S_{vv}^{22} \end{bmatrix} \begin{bmatrix} E_{1h}^+ \\ E_{1v}^+ \\ E_{2h}^+ \\ E_{2v}^+ \end{bmatrix} = \begin{bmatrix} \overline{\overline{S}}_{11} & \overline{\overline{S}}_{12} \\ \overline{\overline{S}}_{21} & \overline{\overline{S}}_{22} \end{bmatrix} \begin{bmatrix} E_{1h}^+ \\ E_{1v}^+ \\ E_{2h}^+ \\ E_{2v}^+ \end{bmatrix} \tag{0.11}$$

We wish to formulate our problem in the coordinate system of the material (xyz) such that

$$\begin{bmatrix} E_{1x}^- \\ E_{1y}^- \\ E_{2x}^- \\ E_{2y}^- \end{bmatrix} = \begin{bmatrix} S_{xx}'^{11} & S_{xy}'^{11} & S_{xx}'^{12} & S_{xy}'^{12} \\ S_{yx}'^{11} & S_{yy}'^{11} & S_{yx}'^{12} & S_{yy}'^{12} \\ S_{xx}'^{21} & S_{xy}'^{21} & S_{xx}'^{22} & S_{xy}'^{22} \\ S_{yx}'^{21} & S_{yy}'^{21} & S_{yx}'^{22} & S_{yy}'^{22} \end{bmatrix} \begin{bmatrix} E_{1x}^+ \\ E_{1y}^+ \\ E_{2x}^+ \\ E_{2y}^+ \end{bmatrix} = \begin{bmatrix} \overline{\overline{S}}'_{11} & \overline{\overline{S}}'_{12} \\ \overline{\overline{S}}'_{21} & \overline{\overline{S}}'_{22} \end{bmatrix} \begin{bmatrix} E_{1x}^+ \\ E_{1y}^+ \\ E_{2x}^+ \\ E_{2y}^+ \end{bmatrix} \tag{0.12}$$

These can be easily related to one another by noting $E_{1,2;x}^{\pm} = E_{1,2;h}^{\pm}\cos\theta$ and $E_{1,2;y}^{\pm} = E_{1,2;v}^{\pm}$

$$\begin{bmatrix} \cos\theta E_{1h}^{-} \\ E_{1v}^{-} \\ \cos\theta E_{2h}^{-} \\ E_{2v}^{-} \end{bmatrix} = \begin{bmatrix} \overline{\overline{S}}'_{11} & \overline{\overline{S}}'_{12} \\ \overline{\overline{S}}'_{21} & \overline{\overline{S}}'_{22} \end{bmatrix} \begin{bmatrix} \cos\theta E_{1h}^{+} \\ E_{1v}^{+} \\ \cos\theta E_{2h}^{+} \\ E_{2v}^{+} \end{bmatrix} \tag{0.13}$$

Which implies

$$\begin{bmatrix} \overline{\overline{S}}'_{11} & \overline{\overline{S}}'_{12} \\ \overline{\overline{S}}'_{21} & \overline{\overline{S}}'_{22} \end{bmatrix} = \begin{bmatrix} \cos\theta & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \overline{\overline{S}}_{11} & \overline{\overline{S}}_{12} \\ \overline{\overline{S}}_{21} & \overline{\overline{S}}_{22} \end{bmatrix} \begin{bmatrix} \sec\theta & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \sec\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{0.14}$$

Let us also define equivalent surface current $\vec{J}^{e} \triangleq -\hat{z}\times\vec{H} \rightarrow \; J_x^e = E_x^e / Z_{\mathrm{TM}}, \;\; J_y^e = E_y^e / Z_{\mathrm{TE}}$ where $Z_{\mathrm{TM}} = \eta_0 \cos\theta$ and $Z_{\mathrm{TE}} = \eta_0 \sec\theta$. From here we can convert into an "ABCD" or "Transmission" matrix which related the fields at the input of the slab to the fields at the output of the slab

$$\begin{bmatrix} \vec{E}_1 \\ \vec{J}_1^e \end{bmatrix} \triangleq \overline{\overline{T}} \begin{bmatrix} \vec{E}_2 \\ \vec{J}_2^e \end{bmatrix} = \begin{bmatrix} \overline{\overline{A}} & \overline{\overline{B}} \\ \overline{\overline{C}} & \overline{\overline{D}} \end{bmatrix} \begin{bmatrix} \vec{E}_2 \\ \vec{J}_2^e \end{bmatrix} \tag{0.15}$$

We can write $\overline{\overline{T}}$ in terms of $\overline{\overline{S}}'$ by noting

$$\vec{E}_{1,2} = \vec{E}_{1,2}^{+} + \vec{E}_{1,2}^{-}, \quad \vec{J}_1^e = \vec{J}_1^{e+} + \vec{J}_1^{e-}, \quad \vec{J}_2^e = -\left(\vec{J}_2^{e+} + \vec{J}_2^{e-}\right) \tag{0.16}$$

and defining $\overleftrightarrow{Z}$ such that $\vec{J}^{e,\pm} = \pm\overleftrightarrow{Z}^{-1}\vec{E}^{\pm} \rightarrow \begin{bmatrix} J_x^{e\pm} \\ J_y^{e\pm} \end{bmatrix} = \pm \begin{bmatrix} \frac{1}{Z_{\mathrm{TM}}} & 0 \\ 0 & \frac{1}{Z_{\mathrm{TE}}} \end{bmatrix} \begin{bmatrix} E_x^{\pm} \\ E_y^{\pm} \end{bmatrix}$. Using these definitions

$$\begin{aligned} &\begin{bmatrix} \vec{E}_1 \\ \vec{J}_1^e \end{bmatrix} = \overline{\overline{T}} \begin{bmatrix} \vec{E}_2 \\ \vec{J}_2^e \end{bmatrix} \rightarrow \begin{bmatrix} \vec{E}_1^{+} + \vec{E}_1^{-} \\ \vec{J}_1^{e+} + \vec{J}_1^{e-} \end{bmatrix} = \overline{\overline{T}} \begin{bmatrix} \vec{E}_2^{+} + \vec{E}_2^{-} \\ \vec{J}_2^{e+} + \vec{J}_2^{e-} \end{bmatrix} \\ &\rightarrow \begin{bmatrix} 1 & 0 \\ 0 & \overleftrightarrow{Z}^{-1} \end{bmatrix} \begin{bmatrix} \vec{E}_1^{+} + \vec{E}_1^{-} \\ \vec{E}_1^{e+} - \vec{E}_1^{e-} \end{bmatrix} = \overline{\overline{T}} \begin{bmatrix} 1 & 0 \\ 0 & -\overleftrightarrow{Z}^{-1} \end{bmatrix} \begin{bmatrix} \vec{E}_2^{+} + \vec{E}_2^{-} \\ \vec{E}_2^{e+} - \vec{E}_2^{e-} \end{bmatrix} \\ &\rightarrow \begin{bmatrix} \vec{E}_1^{+} + S'_{11}\vec{E}_1^{+} + S'_{12}\vec{E}_2^{+} \\ \vec{E}_1^{e+} - S'_{11}\vec{E}_1^{+} - S'_{12}\vec{E}_2^{+} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & \overleftrightarrow{Z} \end{bmatrix} \overline{\overline{T}} \begin{bmatrix} 1 & 0 \\ 0 & -\overleftrightarrow{Z}^{-1} \end{bmatrix} \begin{bmatrix} \vec{E}_2^{+} + S'_{21}\vec{E}_1^{+} + S'_{22}\vec{E}_2^{+} \\ \vec{E}_2^{e+} - S'_{21}\vec{E}_1^{+} - S'_{22}\vec{E}_2^{+} \end{bmatrix} \end{aligned} \tag{0.17}$$

Pulling out the forward electric field components explicitly yields

$$\begin{bmatrix} \mathbf{I} + S'_{11} & S'_{12} \\ \mathbf{I} - S'_{11} & -S'_{12} \end{bmatrix} \begin{bmatrix} \vec{E}_1^{+} \\ \vec{E}_2^{+} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & \overleftrightarrow{Z} \end{bmatrix} \overline{\overline{T}} \begin{bmatrix} 1 & 0 \\ 0 & -\overleftrightarrow{Z}^{-1} \end{bmatrix} \begin{bmatrix} S'_{21} & \mathbf{I} + S'_{22} \\ -S'_{21} & \mathbf{I} - S'_{22} \end{bmatrix} \begin{bmatrix} \vec{E}_1^{+} \\ \vec{E}_2^{+} \end{bmatrix} \tag{0.18}$$

Since the incident waves are arbitrary

$$\overline{\overline{T}} = \begin{bmatrix} 1 & 0 \\ 0 & \overleftrightarrow{Z}^{-1} \end{bmatrix} \begin{bmatrix} \mathbf{I} + S'_{11} & S'_{12} \\ \mathbf{I} - S'_{11} & -S'_{12} \end{bmatrix} \begin{bmatrix} S'_{21} & \mathbf{I} + S'_{22} \\ -S'_{21} & \mathbf{I} - S'_{22} \end{bmatrix}^{-1} \begin{bmatrix} 1 & 0 \\ 0 & -\overleftrightarrow{Z} \end{bmatrix} \tag{0.19}$$

If we define a normalized transmission matrix such that $\begin{bmatrix} \vec{E}_t^L \\ \eta_0 \vec{J}^{e,L} \end{bmatrix} = \overline{\overline{T}}' \begin{bmatrix} \vec{E}_t^R \\ \eta_0 \vec{J}^{e,R} \end{bmatrix}$, then we have

$$\overline{\overline{T}}' = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \sec\theta & 0 \\ 0 & 0 & 0 & \cos\theta \end{bmatrix} \begin{bmatrix} \mathbf{I}+S'_{11} & S'_{12} \\ \mathbf{I}-S'_{11} & -S'_{12} \end{bmatrix} \begin{bmatrix} S'_{21} & \mathbf{I}+S'_{22} \\ -S'_{21} & \mathbf{I}-S'_{22} \end{bmatrix}^{-1} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -\cos\theta & 0 \\ 0 & 0 & 0 & -\sec\theta \end{bmatrix} \tag{0.20}$$

Thus, we now have a matrix from **measurement** which relates the fields at the input and output of the slab. The next step is for us to examine the relation between the ABCD matrix above and the modes within the slab.

3. Converting Material to Theoretical Transmission Matrix

The field at any point within the slab can be described by a superposition of its four modes.

$$\begin{bmatrix} \vec{D}_t \\ \vec{B}_t \end{bmatrix} = \begin{bmatrix} \vec{D}_{t1} \\ \vec{B}_{t1} \end{bmatrix} \alpha_1 + \begin{bmatrix} \vec{D}_{t2} \\ \vec{B}_{t2} \end{bmatrix} \alpha_2 + \begin{bmatrix} \vec{D}_{t3} \\ \vec{B}_{t3} \end{bmatrix} \alpha_3 + \begin{bmatrix} \vec{D}_{t4} \\ \vec{B}_{t4} \end{bmatrix} \alpha_4 = \begin{bmatrix} \vec{D}_{t1} & \vec{D}_{t2} & \vec{D}_{t3} & \vec{D}_{t4} \\ \vec{B}_{t1} & \vec{B}_{t2} & \vec{B}_{t3} & \vec{B}_{t4} \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \end{bmatrix} \tag{0.21}$$

Where the subscript t refers only to the tangential components (x and y) of the D or B. The eigen vectors are obtained by solving the dispersion relation and then enforcing (1.2) and (1.10). We will use the relation in (1.21) to relate the tangential fields at the output to the tangential fields at the input (and thus obtain an ABCD matrix).

At the second (right) side the slab, we have that $\vec{E}_{\tan}^{\text{out,R}} = \vec{E}_{\tan}^{\text{in,R}}$, $\vec{H}_{\tan}^{\text{out,R}} = \vec{H}_{\tan}^{\text{in,R}}$. Recall that $\varepsilon_0 \vec{E}^{\text{in}} = \overleftrightarrow{e}\vec{D}^{\text{in}}$, $\mu_0 \vec{H}^{\text{in}} = \overleftrightarrow{m}\vec{B}^{\text{in}}$. Therefore,

$$\varepsilon_0 \vec{E}^{\text{in}} = \overleftrightarrow{e}\left[\vec{D}_1, \vec{D}_2, \vec{D}_3, \vec{D}_4\right]\overline{\alpha} \tag{0.22}$$

However, since the D field for each mode is solenoidal ($\nabla \cdot D = 0$), $D_{iz} = -\dfrac{p_x}{p_{zi}} D_{ix}$, we have that

$$\begin{aligned} \varepsilon_0 \vec{E}_i = \overleftrightarrow{e}\vec{D}_i = \overleftrightarrow{e} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -\frac{p_x}{p_{zi}} & 0 \end{bmatrix} \vec{D}_{ti} &\rightarrow \varepsilon_0 \vec{E}_{ti} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \overleftrightarrow{e} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -\frac{p_x}{p_{zi}} & 0 \end{bmatrix} \vec{D}_{ti} \\ &\rightarrow \varepsilon_0 \vec{E}_{ti} = \tilde{e}_i \vec{D}_{ti}, \ \tilde{e}_i \triangleq \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \overleftrightarrow{e} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -\frac{p_x}{p_{zi}} & 0 \end{bmatrix} \end{aligned} \tag{0.23}$$

Thus, if we define

$$\tilde{e}_i \triangleq \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \vec{e} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -\frac{p_x}{p_{zi}} & 0 \end{bmatrix}, \quad \tilde{m}_i \triangleq \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \vec{m} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -\frac{p_x}{p_{zi}} & 0 \end{bmatrix} \tag{0.24}$$

Then we can write the tangential electric / magnetic field intensities within the slab as

$$\begin{bmatrix} \vec{E}_t \\ \vec{J}^e \end{bmatrix} = \begin{bmatrix} \vec{E}_{t1} \\ \vec{J}_1^e \end{bmatrix} \alpha_1 + \begin{bmatrix} \vec{E}_{t2} \\ \vec{J}_2^e \end{bmatrix} \alpha_2 + \begin{bmatrix} \vec{E}_{t3} \\ \vec{J}_3^e \end{bmatrix} \alpha_3 + \begin{bmatrix} \vec{E}_{t4} \\ \vec{J}_4^e \end{bmatrix} \alpha_4 = \underbrace{\begin{bmatrix} \vec{E}_{t1} & \vec{E}_{t2} & \vec{E}_{t3} & \vec{E}_{t4} \\ \vec{J}_1^e & \vec{J}_2^e & \vec{J}_3^e & \vec{J}_4^e \end{bmatrix}}_{\overline{\overline{V}}} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \end{bmatrix} \tag{0.25}$$

where $\vec{E}_{ti} = \frac{1}{\varepsilon_0} \tilde{e}_i \vec{D}_{ti}$, $\vec{J}^e = -\hat{z} \times \vec{H}_{ti} = -\hat{z} \times \frac{1}{\mu_0} \tilde{m}_i \vec{B}_{ti}$. Let us assume that the expression above is referenced to the right side of the slab. Each mode on the right side of the slab is related to a mode on the left side of the slab by a propagation constant $k_0 p_{zi} d$, where d is the thickness of the slab. Defining $\lambda_i \triangleq e^{jk_0 p_{zi} d}$ and $\overline{\overline{\Lambda}} \triangleq \mathrm{diag}\left(\lambda_1, \lambda_2, \lambda_3, \lambda_4\right)$, we have

$$\begin{bmatrix} \vec{E}_t^L \\ \vec{J}^{e,L} \end{bmatrix} = \begin{bmatrix} \vec{E}_{t1} \\ \vec{J}_1^e \end{bmatrix} \lambda_1 \alpha_1 + \begin{bmatrix} \vec{E}_{t2} \\ \vec{J}_2^e \end{bmatrix} \lambda_2 \alpha_2 + \begin{bmatrix} \vec{E}_{t3} \\ \vec{J}_3^e \end{bmatrix} \lambda_3 \alpha_3 + \begin{bmatrix} \vec{E}_{t4} \\ \vec{J}_4^e \end{bmatrix} \lambda_4 \alpha_4 = \overline{\overline{V}}\,\overline{\overline{\Lambda}}\,\overline{\alpha} \tag{0.26}$$

But recall that $\begin{bmatrix} \vec{E}_t^R \\ \vec{J}^{e,R} \end{bmatrix} = \overline{\overline{V}}\,\overline{\alpha} \rightarrow \overline{\alpha} = \overline{\overline{V}}^{-1} \begin{bmatrix} \vec{E}_t^R \\ \vec{J}^{e,R} \end{bmatrix}$. Therefore

$$\begin{bmatrix} \vec{E}_t^L \\ \vec{J}^{e,L} \end{bmatrix} = \overline{\overline{V}}\,\overline{\overline{\Lambda}}\,\overline{\overline{V}}^{-1} \begin{bmatrix} \vec{E}_t^R \\ \vec{J}^{e,R} \end{bmatrix} \tag{0.27}$$

Which is the exact same form as (1.15). Therefore, we conclude that

$$\overline{\overline{T}} = \overline{\overline{V}}\,\overline{\overline{\Lambda}}\,\overline{\overline{V}}^{-1} \tag{0.28}$$

This gives us the transmission matrix (ABCD matrix) in terms of **theoretical** material parameters.

4. Algorithmically Computing the T-matrix from Material

Going back again to our eigenvalue problem, we can write this in matrix form as

$$\begin{bmatrix} G_{xx}+1 & G_{xy} \\ G_{yx} & G_{yy}+1 \end{bmatrix}\begin{bmatrix} D_x \\ D_y \end{bmatrix}=0 \tag{0.29}$$

In MATLAB (or other computational tool), the roots of the polynomial in (1.8) can be computed as

$$\overline{p}_z = \text{roots}\left(A_4, A_3, A_2, A_1, A_0\right) \tag{0.30}$$

Where now $\overline{p}_z$ is the array containing all four roots of (1.8). The matrix $\overline{\overline{\Lambda}}$ is now given by

$$\overline{\overline{\Lambda}} = \text{diag}\left(e^{jk_0\overline{p}_z d}\right). \tag{0.31}$$

To compute the modes, we note from (1.29) that

$$\begin{aligned} &\begin{bmatrix} G^i_{xx}+1 & G^i_{xy} \\ G^i_{yx} & G^i_{yy}+1 \end{bmatrix}\begin{bmatrix} D^i_x \\ D^i_y \end{bmatrix}=0 \rightarrow \begin{matrix} \left(G^i_{xx}+1\right)D^i_x = -G^i_{xy}D^i_y \\ -G^i_{yx}D^i_x = \left(G^i_{yy}+1\right)D^i_y \end{matrix} \\ &\rightarrow \left(G^i_{xx}-G^i_{yx}+1\right)D^i_x = \left(G^i_{yy}-G^i_{xy}+1\right)D^i_y \end{aligned} \tag{0.32}$$

which is satisfied when

$$D^i_x = G^i_{yy} - G^i_{xy} + 1, \quad D^i_y = G^i_{xx} - G^i_{yx} + 1. \tag{0.33}$$

This describes the electric flux density of the mode. The magnetic flux density is then

$$\begin{aligned} \vec{B}^i &= \eta_0\left(\vec{p}_i \times \vec{e}\right)\vec{D}^i = \eta_0\vec{p}_i \times \begin{bmatrix} e_x & e_{xy} & e_{zx} \\ e_{xy} & e_y & e_{yz} \\ e_{zx} & e_{yz} & e_z \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -\frac{p_x}{p_{zi}} & 0 \end{bmatrix}\begin{bmatrix} D^i_x \\ D^i_y \end{bmatrix} \\ &= \eta_0\vec{p}_i \times \begin{bmatrix} e_x - \frac{p_x}{p_{zi}}e_{zx}, & e_{xy} \\ e_{xy} - \frac{p_x}{p_{zi}}e_{yz}, & e_y \\ e_{zx} - \frac{p_x}{p_{zi}}e_z, & e_{yz} \end{bmatrix}\begin{bmatrix} D^i_x \\ D^i_y \end{bmatrix} = \eta_0\begin{bmatrix} 0 & -p^i_z & 0 \\ p^i_z & 0 & -p_x \\ 0 & p_x & 0 \end{bmatrix}\begin{bmatrix} e_x - \frac{p_x}{p_{zi}}e_{zx}, & e_{xy} \\ e_{xy} - \frac{p_x}{p_{zi}}e_{yz}, & e_y \\ e_{zx} - \frac{p_x}{p_{zi}}e_z, & e_{yz} \end{bmatrix}\begin{bmatrix} D^i_x \\ D^i_y \end{bmatrix} \\ &= \eta_0\begin{bmatrix} -p^i_z\left(e_{xy} - \frac{p_x}{p_{zi}}e_{yz}\right) & -p^i_z e_y \\ p^i_z\left(e_x - \frac{p_x}{p_{zi}}e_{zx}\right) - p_x\left(e_{zx} - \frac{p_x}{p_{zi}}e_z\right) & p^i_z e_{xy} - p_x e_{yz} \\ p_x\left(e_{xy} - \frac{p_x}{p_{zi}}e_{yz}\right) & p_x e_y \end{bmatrix}\begin{bmatrix} D^i_x \\ D^i_y \end{bmatrix} \end{aligned} \tag{0.34}$$

This confirms (as a sanity check) that $B^i_z = -\frac{p_x}{p_{zi}}B_x$. From the expression above, we can write the x and y components of the magnetic flux density as

$$\begin{aligned}\tfrac{1}{\eta_0}B_x^i &= \left(p_x e_{yz} - p_z^i e_{xy}\right)D_x^i - p_z^i e_y D_y^i \\ \tfrac{1}{\eta_0}B_y^i &= \left[\left(p_z^i e_x - p_x e_{zx}\right) + \tfrac{p_x}{p_{zi}}\left(p_x e_z - p_z^i e_{zx}\right)\right]D_x^i + \left[p_z^i e_{xy} - p_x e_{yz}\right]D_y^i\end{aligned} \tag{0.35}$$

To obtain the electric/magnetic intensities for each mode, we need to compute $\tilde{e}_i$ and $\tilde{m}_i$

$$\begin{aligned}\tilde{e}_i &\triangleq \begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\end{bmatrix}\vec{e}\begin{bmatrix}1 & 0\\ 0 & 1\\ -\frac{p_x}{p_{zi}} & 0\end{bmatrix} = \begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\end{bmatrix}\begin{bmatrix}e_x & e_{xy} & e_{zx}\\ e_{xy} & e_y & e_{yz}\\ e_{zx} & e_{yz} & e_z\end{bmatrix}\begin{bmatrix}1 & 0\\ 0 & 1\\ -\frac{p_x}{p_{zi}} & 0\end{bmatrix} = \begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\end{bmatrix}\begin{bmatrix}e_x - \frac{p_x}{p_{zi}}e_{zx}, & e_{xy}\\ e_{xy} - \frac{p_x}{p_{zi}}e_{yz}, & e_y\\ e_{zx} - \frac{p_x}{p_{zi}}e_z, & e_{yz}\end{bmatrix} \\ &= \begin{bmatrix}e_x - \frac{p_x}{p_{zi}}e_{zx}, & e_{xy}\\ e_{xy} - \frac{p_x}{p_{zi}}e_{yz}, & e_y\end{bmatrix}\end{aligned} \tag{0.36}$$

Therefore

$$\tilde{e}_i = \begin{bmatrix}e_x - \frac{p_x}{p_{zi}}e_{zx}, & e_{xy}\\ e_{xy} - \frac{p_x}{p_{zi}}e_{yz}, & e_y\end{bmatrix}, \quad \tilde{m}_i = \begin{bmatrix}m_x - \frac{p_x}{p_{zi}}m_{zx}, & m_{xy}\\ m_{xy} - \frac{p_x}{p_{zi}}m_{yz}, & m_y\end{bmatrix} \tag{0.37}$$

Thus, the modal quantities for electric field intensity, $\vec{E}_{ti} = \frac{1}{\varepsilon_0}\tilde{e}_i\vec{D}_{ti}$, are

$$\varepsilon_0 E_x^i = \left(e_x - \tfrac{p_x}{p_{zi}}e_{zx}\right)D_x^i + e_{xy}D_y^i, \quad \varepsilon_0 E_y^i = \left(e_{xy} - \tfrac{p_x}{p_{zi}}e_{yz},\right)D_x^i + e_y D_y^i \tag{0.38}$$

The modal quantities for the effective current, $\vec{J}^e = -\hat{z}\times\vec{H}_{ti} = -\hat{z}\times\frac{1}{\mu_0}\tilde{m}_i\vec{B}_{ti}$, are then

$$\begin{aligned}\mu_0\begin{bmatrix}J_x^{ei}\\ J_y^{ei}\end{bmatrix} &= \begin{bmatrix}0 & 1\\ -1 & 0\end{bmatrix}\begin{bmatrix}m_x - \frac{p_x}{p_{zi}}m_{zx}, & m_{xy}\\ m_{xy} - \frac{p_x}{p_{zi}}m_{yz}, & m_y\end{bmatrix}\begin{bmatrix}B_x^i\\ B_y^i\end{bmatrix} = \begin{bmatrix}m_{xy} - \frac{p_x}{p_{zi}}m_{yz}, & m_y\\ \frac{p_x}{p_{zi}}m_{zx} - m_x, & -m_{xy}\end{bmatrix}\begin{bmatrix}B_x^i\\ B_y^i\end{bmatrix} \\ &= \eta_0\begin{bmatrix}m_{xy} - \frac{p_x}{p_{zi}}m_{yz}, & m_y\\ \frac{p_x}{p_{zi}}m_{zx} - m_x, & -m_{xy}\end{bmatrix}\begin{bmatrix}\frac{1}{\eta_0}B_x^i\\ \frac{1}{\eta_0}B_y^i\end{bmatrix}\end{aligned} \tag{0.39}$$

Therefore

$$\tfrac{1}{c_0}J_x^{ei} = \left(m_{xy} - \tfrac{p_x}{p_{zi}}m_{yz}\right)\tfrac{B_x^i}{\eta_0} + m_y\tfrac{B_y^i}{\eta_0}, \quad \tfrac{1}{c_0}J_y^{ei} = \left(\tfrac{p_x}{p_{zi}}m_{zx} - m_x\right)\tfrac{B_x^i}{\eta_0} - m_{xy}\tfrac{B_y^i}{\eta_0} \tag{0.40}$$

Let's rearrange the units a little bit to make it more convenient

$$E_x^i = \left(e_x - \tfrac{p_x}{p_{zi}}e_{zx}\right)\tfrac{D_x^i}{\varepsilon_0} + e_{xy}\tfrac{D_y^i}{\varepsilon_0}, \quad E_y^i = \left(e_{xy} - \tfrac{p_x}{p_{zi}}e_{yz},\right)\tfrac{D_x^i}{\varepsilon_0} + e_y\tfrac{D_y^i}{\varepsilon_0} \tag{0.41}$$

$$\eta_0 J_x^{ei} = \left(m_{xy} - \tfrac{p_x}{p_{zi}}m_{yz}\right)c_0 B_x^i + m_y c_0 B_y^i, \quad \eta_0 J_y^{ei} = \left(\tfrac{p_x}{p_{zi}}m_{zx} - m_x\right)c_0 B_x^i - m_{xy}c_0 B_y^i \tag{0.42}$$

$$
\begin{aligned}
c_0 B_x^i &= \left(p_x e_{yz} - p_z^i e_{xy}\right)\tfrac{D_x^i}{\varepsilon_0} - p_z^i e_y \tfrac{D_y^i}{\varepsilon_0} \\
c_0 B_y^i &= \left[\left(p_z^i e_x - p_x e_{zx}\right) + \tfrac{p_x}{p_{zi}}\left(p_x e_z - p_z^i e_{zx}\right)\right]\tfrac{D_x^i}{\varepsilon_0} + \left[p_z^i e_{xy} - p_x e_{yz}\right]\tfrac{D_y^i}{\varepsilon_0}
\end{aligned}
\tag{0.43}
$$

Since eigenvectors can include an arbitrary constant, we redefine them such that

$$
\tfrac{D_x^i}{\varepsilon_0} = G_{yy}^i - G_{xy}^i + 1, \quad \tfrac{D_y^i}{\varepsilon_0} = G_{xx}^i - G_{yx}^i + 1 \tag{0.44}
$$

Let us define a normalized transmission matrix $\overline{\overline{T}}'$ such that

$$
\begin{bmatrix} \vec{E}_t^L \\ \eta_0 \vec{J}^{e,L} \end{bmatrix} = \overline{\overline{T}}' \begin{bmatrix} \vec{E}_t^R \\ \eta_0 \vec{J}^{e,R} \end{bmatrix} = \overline{\overline{V}}' \overline{\overline{\Lambda}} \overline{\overline{V}}'^{-1} \begin{bmatrix} \vec{E}_t^R \\ \eta_0 \vec{J}^{e,R} \end{bmatrix} \tag{0.45}
$$

This will ultimately be compared to a measured normalized transmission matrix computed as

$$
\overline{\overline{T}}' = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \sec\theta & 0 \\ 0 & 0 & 0 & \cos\theta \end{bmatrix} \begin{bmatrix} \mathbf{I} + S_{11}' & S_{12}' \\ \mathbf{I} - S_{11}' & -S_{12}' \end{bmatrix} \begin{bmatrix} S_{21}' & \mathbf{I} + S_{22}' \\ S_{21}' & -\mathbf{I} + S_{22}' \end{bmatrix}^{-1} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \cos\theta & 0 \\ 0 & 0 & 0 & \sec\theta \end{bmatrix} \tag{0.46}
$$

5. Summaries

Theoretical to T Matrix

Given material parameters, we can build a T matrix by first computing the A coefficients in (1.8) and the G matrix in (1.3) through (1.6) and then computing (in this order)

$$\overline{p}_z = \text{roots}\left(A_4, A_3, A_2, A_1, A_0\right) \tag{0.47}$$

$$\overline{\overline{\Lambda}} = \text{diag}\left(e^{jk_0\overline{p}_z d}\right) \tag{0.48}$$

$$\tfrac{D_x^i}{\varepsilon_0} = G_{yy}^i - G_{xy}^i + 1, \quad \tfrac{D_y^i}{\varepsilon_0} = G_{xx}^i - G_{yx}^i + 1 \tag{0.49}$$

$$E_x^i = \left(e_x - \tfrac{p_x}{p_{zi}} e_{zx}\right)\tfrac{D_x^i}{\varepsilon_0} + e_{xy}\tfrac{D_y^i}{\varepsilon_0}, \quad E_y^i = \left(e_{xy} - \tfrac{p_x}{p_{zi}} e_{yz},\right)\tfrac{D_x^i}{\varepsilon_0} + e_y\tfrac{D_y^i}{\varepsilon_0} \tag{0.50}$$

$$\begin{aligned} c_0 B_x^i &= \left(p_x e_{yz} - p_z^i e_{xy}\right)\tfrac{D_x^i}{\varepsilon_0} - p_z^i e_y \tfrac{D_y^i}{\varepsilon_0} \\ c_0 B_y^i &= \left[\left(p_z^i e_x - p_x e_{zx}\right) + \tfrac{p_x}{p_{zi}}\left(p_x e_z - p_z^i e_{zx}\right)\right]\tfrac{D_x^i}{\varepsilon_0} + \left[p_z^i e_{xy} - p_x e_{yz}\right]\tfrac{D_y^i}{\varepsilon_0} \end{aligned} \tag{0.51}$$

$$\eta_0 J_x^{ei} = \left(m_{xy} - \tfrac{p_x}{p_{zi}} m_{yz}\right) c_0 B_x^i + m_y c_0 B_y^i, \quad \eta_0 J_y^{ei} = \left(\tfrac{p_x}{p_{zi}} m_{zx} - m_x\right) c_0 B_x^i - m_{xy} c_0 B_y^i \tag{0.52}$$

$$\overline{\overline{V}}' \triangleq \begin{bmatrix} E_x^1 & E_x^2 & E_x^3 & E_x^4 \\ E_y^1 & E_y^2 & E_y^3 & E_y^4 \\ \eta_0 J_x^{e1} & \eta_0 J_x^{e2} & \eta_0 J_x^{e3} & \eta_0 J_x^{e4} \\ \eta_0 J_y^{e1} & \eta_0 J_y^{e2} & \eta_0 J_y^{e3} & \eta_0 J_y^{e4} \end{bmatrix} \tag{0.53}$$

$$\overline{\overline{T}}' = \overline{\overline{V}}' \overline{\overline{\Lambda}} \overline{\overline{V}}'^{-1} \tag{0.54}$$

Summary – Measurement to T Matrix

After taking measurements with the SPAM system in the h-v coordinate system we perform computations in the following order to obtain the normalized transmission matrix

$$\begin{bmatrix} \overline{\overline{S}}'_{11} & \overline{\overline{S}}'_{12} \\ \overline{\overline{S}}'_{21} & \overline{\overline{S}}'_{22} \end{bmatrix} = \begin{bmatrix} \cos\theta & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \overline{\overline{S}}_{11} & \overline{\overline{S}}_{12} \\ \overline{\overline{S}}_{21} & \overline{\overline{S}}_{22} \end{bmatrix} \begin{bmatrix} \sec\theta & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \sec\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{0.55}$$

$$\overline{\overline{T}}' = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \sec\theta & 0 \\ 0 & 0 & 0 & \cos\theta \end{bmatrix} \begin{bmatrix} \mathbf{I} + S'_{11} & S'_{12} \\ \mathbf{I} - S'_{11} & -S'_{12} \end{bmatrix} \begin{bmatrix} S'_{21} & \mathbf{I} + S'_{22} \\ S'_{21} & -\mathbf{I} + S'_{22} \end{bmatrix}^{-1} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \cos\theta & 0 \\ 0 & 0 & 0 & \sec\theta \end{bmatrix} \tag{0.56}$$

6. Simulated Test Cases

An HFSS simulation will provide generalized (power-wave) scattering parameters. To convert from these to electric-field scattering parameters:

$$\begin{bmatrix} \bar{\bar{S}}'_{11} & \bar{\bar{S}}'_{12} \\ \bar{\bar{S}}'_{21} & \bar{\bar{S}}'_{22} \end{bmatrix} = \begin{bmatrix} \sqrt{2Z_{\mathrm{TM}}} & 0 & 0 & 0 \\ 0 & \sqrt{2Z_{\mathrm{TE}}} & 0 & 0 \\ 0 & 0 & \sqrt{2Z_{\mathrm{TM}}} & 0 \\ 0 & 0 & 0 & \sqrt{2Z_{\mathrm{TE}}} \end{bmatrix} \begin{bmatrix} \bar{\bar{S}}_{11} & \bar{\bar{S}}_{12} \\ \bar{\bar{S}}_{21} & \bar{\bar{S}}_{22} \end{bmatrix} \begin{bmatrix} \sqrt{\frac{1}{2Z_{\mathrm{TM}}}} & 0 & 0 & 0 \\ 0 & \sqrt{\frac{1}{2Z_{\mathrm{TE}}}} & 0 & 0 \\ 0 & 0 & \sqrt{\frac{1}{2Z_{\mathrm{TM}}}} & 0 \\ 0 & 0 & 0 & \sqrt{\frac{1}{2Z_{\mathrm{TE}}}} \end{bmatrix} \quad (0.57)$$

$$\bar{\bar{T}}' = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \sec\theta & 0 \\ 0 & 0 & 0 & \cos\theta \end{bmatrix} \begin{bmatrix} \mathbf{I} + S'_{11} & S'_{12} \\ \mathbf{I} - S'_{11} & -S'_{12} \end{bmatrix} \begin{bmatrix} S'_{21} & \mathbf{I} + S'_{22} \\ S'_{21} & -\mathbf{I} + S'_{22} \end{bmatrix}^{-1} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \cos\theta & 0 \\ 0 & 0 & 0 & \sec\theta \end{bmatrix} \quad (0.58)$$

Five materials slabs have been simulated so far and converted into scattering parameters that would hypothetically be given by the SPAM system. Each material is 60 mil thick and illuminated over angles between 2 and 88 degrees in steps of 2 degrees. I have already taken the HFSS files and converted them into ***S'*** using (1.57). The files for each material sample are named with the prefix SPAM_Scat_Mat_*. The material parameters corresponding to each file suffix are given below

$$\text{er2_mur3} \leftrightarrow \bar{\bar{\varepsilon}}_r = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}, \bar{\bar{\mu}}_r = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}, \qquad \text{er225_mur333} \leftrightarrow \bar{\bar{\varepsilon}}_r = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{bmatrix}, \bar{\bar{\mu}}_r = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$\text{er225_mur343} \leftrightarrow \bar{\bar{\varepsilon}}_r = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{bmatrix}, \bar{\bar{\mu}}_r = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 3 \end{bmatrix}, \qquad \text{er201205_mur343} \leftrightarrow \bar{\bar{\varepsilon}}_r = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 5 \end{bmatrix}, \bar{\bar{\mu}}_r = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$\text{er201205_allj_mur343_allj} \leftrightarrow \bar{\bar{\varepsilon}}_r = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 5 \end{bmatrix}(1-j), \bar{\bar{\mu}}_r = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 3 \end{bmatrix}(1-j)$$

I have written a code which compares the normalized transmission matrices to their theoretically computed counterparts. There is something off between the Transmission matrices (which I believe originates from disagreement in the eigenvector computation since the eigenvalues seem to agree quite well aside from a mysterious sign).